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DEPARTMENT OF INFORMATION SYSTEMS

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EXAMINER/S : P Monyeki
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INSTRUCTIONS/REQUIREMENTS:-

1. Answer all the questions.
2. Adhere to the time limit
3. Write neatly and legibly

Do not turn the page until permission is given

Question One **[30 Marks]**

Question one consists of two sections. Section A and B.

Section: A (Multiple-choice Questions)

Choose the most appropriate answer for each of the questions: e.g. 2.1 E

1.1. A group is switching to Scrum. They employed a project coordinator whose role involved creating a supportive work environment for all team members involved in projects. Following the transition, who will be responsible for the team? [1X2]

A. Developer

B. Product Owner

C. Project manager

D. Scrum Master

1.2 Which of the following option is not considered as an appropriate technique for estimating velocity? [1X2]

A. Use historical averages

B. Run 1-2 iterations

C. Predict

D. Forecast

1.3 Which of the following is completed in the transition phase of the Unified Process (UP)? [1X2]

A. Complete beta test

B. Resolve high risk

C. Implement the core functions

D. Make the plans to deploy

1.4 Which one of the following is NOT correct about the important stage of product development, that is to do estimation? [1X2]

A. estimating the [resources, time, size] required to develop a [user story, feature, or requirements]

B. Based on size, not duration

C. Compares relative effort

D. Estimations are done at the sprint backlog level, not at the product backlog level.

1.5 Which of the following options does not adhere to the purpose of the Sprint Retrospective Meeting? [1X2]

A. Discuss the impediments raised by the development team during the last sprint and plan for implementing improvements.

B. Refinement of epics nominated by the product owner for the next couple of sprints to promote reliable release planning

C. Discuss the interaction within the Scrum Team and agree upon measures to improve the collaboration.

D. All the mentioned

1.6 _____is an approach to the SDLC (System development life cycle) that plans the project in advance and then progresses according to the plan [1X2]

A. An adaptive approach

B. A Predictive approach

C. An Agile approach

D. A project management approach

1.7 Which of the following is the description of a burndown graph? [1X2]

A. A Sprint plan which is burned to celebrate the successful completion of a Sprint

B. The trend of work remaining across time in a Sprint, a release, or a product

C. The trend of work completed across time in a Sprint, a release, or a product

D. A graph used to measure the burnout of the Development Team due to the fast-paced nature of a Sprint

1.8 Unified Process (UP) disciplines are involved in each iteration, which is typically planned to span for ____ week/s. [1X2]

A. One

B. two

C. four

D. eight

1.9 _____is an agile project management tool that provides user with simple, natural language explanations of one or more features written from the end-user perspective. [1X2]

A. User story

B. Scrum

C. Sprint

D. Model

1.10 Which of the following is **not** the statement to define the database structure of schema? [1X2]

A. Create

B. Insert

C. Drop

D. Rename

Section: B (True/False questions)

Choose the most appropriate answer for each of the questions: e.g. 2.1 E

1.11 A project that has high technical risk should use the predictive approach. [1X2]

A. True

B. False

1.12 One of the benefits from Scrum is that the Development Team doesn't have to write detailed specifications anymore. [1X2]

A. True

B. False

1.13 In incremental development, an increment is always completed in each iteration. [1X2]

A. . True

B. False

1.14 Extreme Programming (XP) recommends that system code be written before test code. [1X2]

A. True

B. False

1.15 As a scrum master you would like to come up with time estimate required to accomplish specific piece of work using velocity. [1X2]

A. True

B. False

Question Two **[10 Marks]**

Match column A with Column B. Write only the question number and the corresponding letter. e.g., 2.1 B

Column A	Column B
2.1 Tools	A. Develop an approximate version of the system, make the business case, define scope, and produce rough estimates for cost and schedule.
2.2 Inception	B. Provides guidelines for every step of system development: What to do when, why and how
2.3 Confirmation	C. is something tangible that works and which we could potentially deliver, if it was enough to provide business benefit.
2.4 Methodology	D. Software applications that assist developers in creating models or other components required for a project
2.5 feature	E. Conditions that need to satisfy to ensure that the story meets all requirements

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Question Three **[10 Marks]**

Fill in the missing word. Write the question number and the appropriate answer.

3.1 Adaptive is the SDLC approach that is implemented when requirements are uncertain. [1X2]

3.2 The objective of construction is to iteratively implement the remaining lower risk, predictable, and easier elements and prepare for deployment. [1X2]

3.3 use case is the formal description of a starting state, one or more events to which the software must respond, and the expected response or ending state. [1X2]

3.4 stub is a method or class developed for unit testing that simulates the behaviour of a method invoked that hasn't yet been written. [1X2]

3.5 ..... system requirements description and architecture to help maintenance and upgrade of the system. [1X2] reverse engineering

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Question Four **[15 Marks]**

Read the following scenario and answer the following questions given below:

Mrs. Naidoo works as a system analyst in the Nang Antap Company. She has been asked to upgrade the company's Information System to solve various problems faced by the users as well as to strengthen the company in line with the present situation. One thing that Mrs. Naidoo needs to do is to construct interviews among users of Nang Antap Company. She completes her tasks by finishing all phases in System Development and her boss is very satisfied with her work.

Based on the above scenario, answer the following questions:

4.1 Based on the underlined statement, state the phase involved in System Development. [1X2]

4.2 Describe another step that can be done by Mrs. Naidoo in the phase provided in question 4.1. [3X1]

4.3 Outline six (6) phases of the traditional predictive systems development life cycle (SDLC). [6X1]

4.4 1 List and discuss two general approaches to SDLC. [2X2]

Question Five **[15 Marks]**

You work as a cybersecurity analysist for a multinational IT company in South Africa. One day, you receive an email from an employee who claims to have received a suspicious email with a subject line that suggests it contains confidential information. The employee is concerned that it might be a phishing attempt. You investigate further and discover that the email does indeed contain a malicious attachment.

Based on the above scenario, answer the following questions:

5.1 Describe two immediate actions you would take to handle this incident and mitigate potential risks.

[2X2]

5.2 Describe how trojan malware works. [1X1]

5.3 There are two general approaches to planning sprints: velocity-driven planning and commitment-driven planning.

5.3.1Discuss each approach and provide one advantage and one disadvantage. [3X2]

5.3.2 After the scrum team have met with the product owner who provided them with requirements, they conducted a sprint-planning meeting. List and explain any **two artefacts** that the team must have when the sprint planning meeting ends (adjourned). [2X2]

Question Six **[20 Marks]**

6. The following statements describe the process for adding a vehicle to an existing insurance policy.

Customer provides information on car to be added, selects coverage, and identifies drivers of the new car. System updates the policy.

Clerk enters policy number. System finds policy and displays details. Clerk verifies policy is current. System validates that car has known standard. Clerk enters car identification information. Clerk enters each type of coverage customer requests, including deductibles and coverage amount. System validates coverage requests and adds to policy. System validates combination coverages. Clerk indicates end of session. System updates policy, calculates new premium, and prints new statement.

Based on the above scenario. Draw an activity diagram with swim lanes to show all processes.